INCH POUND
MIL-L-52998A(ME)
21 October 1988
SUPERSEDING
MIL-L-52998(ME)
4 December 1981

MILITARY SPECIFICATION

LIGHT, MARKER, BLINKING

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-contained, battery-operated, flashing orange light.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated - Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6230
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MILITARY

MIL-B-18	- Batteries, Dry.
MIL-B-18/9	- Batteries, Dry, BA-30.
MIL-P-116	- Preservation, Methods of.
MIL-B-49030	- Batteries, Dry (Alkaline).
MIL-B-49030/1	- Batteries, Dry, BA-3030/U.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DRAWINGS

ME

TA13221E7385	- Light, Marker, Blinking.
13221E7392	- Box, Light.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C778 - Standard Sand, Specification for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The light, marker, blinking (hereinafter referred to as the light) shall consist of a flasher assembly, case, gasket, and battery contact spring in accordance with Belvoir Drawing TA13221E7385. The light shall be activated by tightening the flasher assembly to the gasket to provide a leakproof seal without significantly deforming the gasket. The light shall be deactivated by loosening the flasher assembly from the case but not removing it; the flasher assembly shall remain attached to the case when the light is turned off although deactivation shall coincide with loss of seal.

3.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without the prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the light and light box assembly. Any data (e.g., shop drawings, layouts, flow sheets, and processing procedures) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.3 Components. Components of the light shall be fabricated of materials and in dimensions as specified on the applicable drawings.

3.4 Materials. Materials shall be as specified on the applicable drawings. The materials shall be free from all defects and imperfections that may affect the quality and reliability of the finished product.

3.4.1 Material deterioration prevention and control. The lights shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the lights may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

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3.4.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the lights may be newly fabricated from recovered materials to the maximum extent practicable, provided the lights produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the lights.

3.5 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.6 Interchangeability. The flasher assembly, battery contact spring, case, and gaskets shall be interchangeable between lights. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.7 Angle of visibility. The light emitted from the flasher assembly shall be visible through an arc of 180 degree ±10 degree horizontally and through an angle of 30 degree (+10 degree, -2 degree) above and below the horizontal axis through the electrodes when the light is held vertically. The light source shall not be visible from directly overhead.

3.8 Compatibility with cells. The light shall be physically and electrically compatible with (type BA30) cells conforming to MIL-B-18/9 and (type BA3030U) cells conforming to MIL-B-49030/1.

3.8.1 Government furnished property. When specified (see 6.2), either of the following items, or both, in the quantities indicated, will be furnished by the Government.

Item No.	Description	Identification	Quantity
1	Cell, dry, single voltage, 1.5 volts, MIL-B-18/9, type BA30	6135-00-120-1020	AR
2	Cell, dry, single voltage, 1.5 volts, MIL-B-49030/1, type BA3030U	6135-00-930-0030	AR

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If a light should fail one of the tests specified in Section 4 and the contractor believes the failure resulted from one of the above cells, the contractor may conduct such further tests as desired, at his expense, to determine which component failed.

3.9 Weight. The light without cell shall weigh no more than 0.36 pound.

3.10 Identification marking. Each light shall be permanently and distinctively marked with the letters US (at least 0.2 inch high), the name of the manufacturer, and the part number.

3.11 Workmanship. The light, including all parts and accessories, shall be fabricated and finished to insure freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, fillet radii, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, wiring, and riveting; alinement of parts; and tightness of assembly components. The lens shall be free of easily visible embedded particles and bubbles.

3.12 Light characteristics. The contractor shall procure or fabricate measurement equipment to determine the effective photopic luminous intensity of the lights. All measurements shall be made in the same physical units and to the same scale factor, enabling direct numerical comparison of measured intensities, current drains, and flash rates (or periods).

3.12.1 Luminous intensity. As determined by the measurement equipment specified in 3.12, the effective photopic luminous intensity of each production light shall be within 2.5 standard deviations of the average intensity of the 30 model lights as determined in 4.4.3.2.2.

3.12.2 Environmental conditions.

3.12.2.1 Room temperature flash rate. The lights shall pass room temperature flash rate test described in 4.4.3.10.

3.12.2.2 High temperature. The lights shall pass the high temperature test described in 4.4.3.3.

3.12.2.3 Humidity. The lights shall pass the humidity test described in 4.4.3.4.

3.12.2.4 Cold temperature. The lights shall pass the cold temperature test described in 4.4.3.5.

3.12.2.5 Intermediate cold. The lights shall pass the intermediate cold test described in 4.4.3.6.

3.12.2.6 Rough use. The light shall pass the drop test as specified in 4.4.3.7.

3.12.2.7 Leakage (immersion). The lights shall be leakproof when tested as specified in 4.4.3.8.1 and 4.4.3.8.2.

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3.12.2.7.1 Flasher assembly seal test. The flasher assemblies, removed from the lights, shall be leakproof when tested as specified in 4.4.3.8.3 and 4.4.3.8.4.

3.12.3 Low-voltage operation. The light shall pass the low-voltage tests described in 4.4.3.12 and 4.4.3.13.

3.12.4 Reliability. The specified reliability for a 360-hour mission shall be 0.99 for the light when tested as specified in 4.4.3.11.

3.12.5 Maintainability. The lights shall pass the cell replacement test described in 4.4.3.9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.5).

4.3 First article inspection. The Government shall randomly select a number of lights (see 6.2), of the first article models being produced by production tooling. These lights will be subjected to the first article inspection examination by the contractor and witnessed by the Government. The lights shall be examined for the characteristics marked "X" in column 2 of table I of 4.4.2.1 and subjected to the tests marked "X" in column 2 of table II of 4.4.2.2 to determine conformance to the requirements of this specification. Acceptance of a first article light shall not exclude the remaining lights from the quality performance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the first article test, the Government

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reserves the right to conduct any or all other tests contained in this specification as part of the first article test, and failure of such additional tests shall have the same effect as failure of those tests specified as first article tests.

4.3.1 Failure of inspection. Acceptance and rejection criteria are specified in 4.4. Failure of first article lights to meet examination criteria marked "X" in column 2 of table I of 4.4.2.1 or test criteria marked "X" in column 2 of table II of 4.4.2.2 shall be cause for rejection of the first article light. Further, any failure shall be cause for the Government to refuse to accept any production lights until the contractor has provided acceptable evidence that action has been taken to eliminate the defects. Correction of such defects shall be accomplished by the contractor at no cost to the Government on lights previously accepted and produced under the contract. Defects found as a result of the first article inspection will be considered prima facie evidence that all lights accepted prior to the completion of first article inspection are similarly defective unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.4 Quality conformance inspection.

4.4.1 Definitions.

4.4.1.1 Unit of product. The unit of product is one light.

4.4.1.2 Inspection lot. An inspection lot is a homogeneous number of units of product produced to the same specifications by the same procedures through the same process at the same location at essentially the same time.

4.4.1.3 Sampling. Sampling shall be in accordance with MIL-STD-105, level II.

4.4.2 Inspection procedures.

4.4.2.1 Examination. Samples selected in accordance with 4.4.1.3 shall be examined for the defects marked "X" in column 2 of table I. Acceptable quality level (AQL) shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

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TABLE I. Examination schedule.

Column 1	Column 2 Type of Examination		Column 3	Column 4
Number	First Article	Quality Conformance	Characteristic	Requirements Paragraph
Major				
101	X	X	Dimension not as specified.	3.1
102	X	X	Part missing or damaged.	3.1
103	X	X	Materials or components not as specified.	3.3, 3.4
104	X	X	Retroreflective material not as specified.	3.4
105	X	X	Visibility angles not as specified.	3.7
106	X	X	Flasher not permanently fixed to lens.	3.2
107	X	-	Incompatible with cells.	3.8
108	X	-	Weight more than 0.36 lb.	3.9
109	X	X	Light does not activate by tightening lens to case and gasket.	3.1
110	X	X	Case does not have required number of threads.	3.2
111	X	X	Flahser assembly not attached to case after deactivation.	3.1
112	X	X	Gasket installed improperly.	3.3
113	X	X	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.4.1
114	X	X	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.4.2
115	X	X	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.4.3
116	X	X	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the lights.	3.4.4
Minor				
201	X	X	Identification marking missing, illegible, or incorrect.	3.10
202	X	X	Workmanship not as specified.	3.11

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4.4.2.2 Tests. Samples selected in accordance with 4.4.1.3 shall be subjected to the tests marked "X" in column 2 of table II. The order shall be as shown in column I of table II. Failure of the sample to meet an AQL of 1.0 percent defective shall be cause to reject the lot.

TABLE II. Test schedule.

Column 1	Column 2 Type of Test		Column 3	Column 4	Column 5
Order of Tests	First Article	Quality Conformance	Test	Test Paragraph	Requirements Paragraph
1	X	-	Burn-in.	4.4.3.1	3.12.1
2	X	X	Luminous intensity.	4.4.3.2	3.12.1
3	X	X	High temperature.	4.4.3.3	3.12.2.2
4	**	-	Humidity.	4.4.3.4	3.12.2.3
5	**	X	Cold temperature.	4.4.3.5	3.12.2.4
6	**	-	Intermediate cold Temperature.	4.4.3.6	3.12.2.5
7	X	-	Rough use.	4.4.3.7	3.12.2.6
8	X	X	Leakage (immersion).	4.4.3.8	3.12.2.7
*	X	X	Cell replacement.	4.4.3.9	3.12.5
*	X	X	Room temperature flash.	4.4.3.10	3.12.2.1
*	X	***	Reliability.	4.4.3.11	3.12.4
*	X	X	Low-voltage.	4.4.3.12& 4.4.3.13	3.12.3

* Order not specified.
** Tests will be performed during reliability testing.
*** See 4.4.3.11.3.

4.4.3 Test description.

4.4.3.1 Burn-in test. The flasher assembly shall be connected to a calibrated power supply set at 1.5 volts ± .02 volt and a record of the effective luminous intensity obtained. The effective luminous intensity shall be measured as described in 4.4.3.2.2. Install a cell and allow the light to operate 360 hours. (Note: The 360-hour operating requirement can be satisfied by conducting either the tests outlined in 4.4.3.4.1 or 4.4.3.6.2.) At the end of 360 hours of operation, reconnect the flasher assembly to the calibrated power supply at 1.5 volts ± .02 volt and repeat the measurement of the effective luminous intensity in accordance with 4.4.3.2.2.

4.4.3.1.1 Failure criteria. The light shall be rejected if the effective luminous intensity after 360 hours of operation differs by greater than 2.5 standard deviations from the standard intensity specified in 4.4.3.2.2.

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4.4.3.2 Luminous intensity test.

4.4.3.2.1 Test conditions. The luminous intensity test shall be made at a temperature of 73 ± 5 °F with a calibrated power supply set at 1.5 volts ± .02 volt.

4.4.3.2.2 Standard effective luminous intensity. Using measurement equipment procured or fabricated for this purpose the contractor shall determine the effective photopic luminous intensity (see 6.6) for each of 30 lights, furnished on loan by the Government. The average effective photopic luminous intensity shall be determined and will be used as a standard for which individual contract production lights will be compared. The measuring equipment shall be checked to determine that all measurements are relative.

4.4.3.2.3 Effective luminous intensity. The contractor shall determine the effective luminous intensity of each production light being tested (see 4.4.3.2.2).

4.4.3.2.4 Failure criteria. Any light tested with an effective luminous intensity less than the standard effective luminous intensity, decreased by 2.5 standard deviations, as determined in accordance with 4.4.3.2.2 shall constitute failure of this test.

4.4.3.3 High temperature test.

4.4.3.3.1 Test procedure. The light, without cell, shall be tested in accordance with MIL-STD-810, method 501.2, procedure II. The operating temperature shall be 135 ±5 °F. A fresh cell is inserted prior to step 10 and the light operated for not less than 2 hours. Within 2 hours after returning the light to ambient temperature, the flash rate shall be measured and each light shall be examined.

4.4.3.3.2 Failure criteria. The light shall fail if it shows evidence of or contains any of the following:

a. Discoloration, cracking, crazing, or distortion of the lens.
b. Cracking, distortion, or bulging of the light case.
c. Condensation of moisture on inside of light.
d. Distortion, softening, cracking, or embrittlement of the gasket.
e. Failure to flash between 70 and 110 times per minute.
f. Failure to flash with luminous intensity within 2.5 standard deviations of the standard intensity specified in 4.4.3.2.2.

4.4.3.4 Humidity test.

4.4.3.4.1 Test procedure. Insert a fresh cell (MIL-B-49030/1, type BA3030U) and activate light. Test in accordance with MIL-STD-810, method 507.2, procedure II, except six continuous 48-hour cycles shall be performed in step 6. The

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maximum temperature during this test shall be 125 ± 5 °F. At the completion of step 7, the light shall be returned to standard ambient temperature for 1 hour and the effective luminous intensity determined as specified in 4.4.3.2.2 without cell replacement.

4.4.3.4.2 Failure criteria. The light shall fail if it shows evidence of or contains any of the following:

a. Discoloration, cracking, crazing, or distortion of the lens.
b. Cracking, distortion, or bulging of the light case.
c. Condensation of moisture on inside of light.
d. Distortion, softening, cracking, or embrittlement of the gasket.
e. Failure to flash between 70 and 110 times per minute.
f. Failure to flash with luminous intensity within 2.5 standard deviations of the standard intensity specified in 4.4.3.2.2.

4.4.3.5 Cold temperature test.

4.4.3.5.1 Test procedure. The light shall have a fresh cell (MIL-B-49030/1, type BA3030U) inserted. The light shall be activated and stored at a temperature of -35 ± 3 °F for 3 hours. At the end of this time, at a temperature of -35 ± 3 °F, the flash rate shall be measured.

4.4.3.5.2 Failure criteria. The light shall fail if it does not flash within a range of 50 to 110 times per minute.

4.4.3.6 Intermediate cold temperature test.

4.4.3.6.1 Test conditions. If the test 4.4.3.5 is conducted, the cells and lights utilized in 4.4.3.5 shall be used to conduct the test specified herein. If the test in 4.4.3.5 is not conducted, a light with a fresh cell (MIL-B-49030/1, type BA3030U) shall be used to conduct the test specified herein.

4.4.3.6.2 Test procedure. The light with cell shall be activated and stored at the following temperature for the times shown:

Temperature (°F)	Time (hours)*
-35 ± 3	3 (not required if previously tested in 4.4.3.5)
-25 ± 3	9
- 4 ± 3	108
-25 ± 3	1
- 4 ± 3	118
-25 ± 3	1
- 4 ± 3	118
-35 ± 3	1
- 4 ± 3	1

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* The time required for transition between temperatures is included in the times listed. The transition rate between temperatures shall be equal to or greater than 1 degree per minute. At the end of this time and at a temperature of -4 ± 3 °F, the flash rate of the light shall be measured.

4.4.3.6.3 Failure criteria. The light shall fail if it does not flash within the range of 25 to 110 flashes per minute.

4.4.3.7 Rough use.

4.4.3.7.1 Test procedure. The light shall be dropped, complete with cell, from a height of 12 feet, at a temperature of 73 ± 5 °F, onto the center of a bed of loose, dry sand, to constitute a free fall. The sand used shall be standard sand in accordance with ASTM C778. The sand bed shall be at least 2 feet by 2 feet by 6 inches deep and the exposed surface of the sand shall be smoothed to a horizontal plane prior to each drop. The light shall be dropped so that it strikes 10 times in an approximate horizontal position (5 of which will be so oriented as to have the front of the bulb face the ground), 10 times in a vertical position upon the dome of the light, and 10 times in a vertical position upon the base of the light. Parts shall not be tightened once the test has begun.

4.4.3.7.2 Failure criteria. Failure of the light to flash as specified in 3.12.1 or 3.12.2.1 or any cracking of the lens, bulb, case, base, or post, or the detachment of any of these, shall constitute a failure. Three or more failures accumulated during rough use test shall be cause for rejection.

4.4.3.8 Leakage (immersion) tests.

4.4.3.8.1 Light seal test procedure. The light, with cell, shall be tested in accordance with MIL-STD-810, method 512.2, procedure I, at air pressure above the water of 19 inches of mercury, increased to 20 inches after 1 minute.

4.4.3.8.2 Failure criteria. The light shall fail if there is any sign of leakage or condensation on the inside of the flasher assembly or battery compartment.

4.4.3.8.3 Flasher assembly seal test procedure. The flasher assembly, removed from the light, shall be similarly tested in accordance with MIL-STD-810, method 512.2, procedure I, at air pressure above the liquid of 19 inches of mercury, increased to 20 inches of mercury after 1 minute.

4.4.3.8.4 Failure criteria. The light shall fail if there is any sign of leakage or condensation on the inside of the dome.

4.4.3.9 Cell replacement test.

4.4.3.9.1 Test procedure. Insure that light being subjected to this test has the flasher assembly fully screwed onto the case and the light is activated. Measure the total time required to:

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a. Remove the flasher assembly from the light.
b. Remove the cell and install another cell.
c. Reinstall the flasher assembly on the case.

Determine the average time required to accomplish a, b, and c.

4.4.3.9.2 Failure criteria. The light shall fail if the average time required for cell replacement exceeds 15 seconds.

4.4.3.10 Room temperature flash rate test.

4.4.3.10.1 Test procedure. The light shall have a fresh MIL-B-18/9 (type BA30) cell inserted. The light shall be allowed to flash in a room at 75 ± 10 °F.

4.4.3.10.2 Failure criteria. The light shall fail if the flash rate is not between 70 and 110 times per minute.

4.4.3.11 Reliability test. When specified (see 6.2), the following reliability test shall be performed.

4.4.3.11.1 Test procedure. Demonstration of the minimum reliability as specified in 3.12.4 shall be conducted in accordance with the following test procedure for first article lots. At least 200 lights shall be operated for a continuous period of 360 hours as specified in 4.4.3.6. Immediately following this test, fresh cells (MIL-B-49030/1, type BA3030U) shall be inserted and the same lights shall be operated for a second continuous period of 360 hours as specified in 4.4.3.4.

4.4.3.11.2 Failure criteria. At the completion of the first 360 hours of light operation, as specified in 4.4.3.6.2, failure criteria 4.4.3.6.3 will apply. At the completion of the second 360 hours of operation, as specified in 4.4.3.4.1 failure criteria 4.4.3.4.2 will apply. Three or more failures accumulated during the test will constitute failure of the reliability test.

4.4.3.11.3 Reliability check test. The reliability check test shall consist of operating 200 light units under ambient conditions for a period of 360 hours continuously, allowing five or fewer failures (see 6.2). Every quarter year of production shall have this reliability check test. Failure of this test shall be cause for refusal by the Government to continue acceptance of production lights until evidence has been provided by the contractor that corrective action has been taken to correct the deficiency. Any deficiency found as a result of the check test will be considered prima facie evidence that all lights provisionally accepted since the last check test are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government.

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4.4.3.12 Low-voltage flash rate test. The light shall operate at a flash rate no slower than 25 times per minute when activated by low voltage (0.75 V), as would be the case for a small fraction of new dry cells, or for activation of average dry cells in cold temperatures.

4.4.3.13 Low-voltage current drain test. The light shall operate at a current drain no greater than 6 mA when activated by low voltage (0.75 V).

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection.

4.5.1.1 Unit of product. For purposes of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

113. Lights not preserved as specified.
114. Lights not packed as specified for level A, B, or C.
115. Lights not palletized as specified when required.
116. Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be level A (see 6.5).

5.1.1 Lights procured as sets for Hand Emplaceable Minefield Marking Set (HEMMS). Seventy-two lights shall be preserved together in accordance with MIL-P-116, method IA-15. The container and partitions shall be fabricated in accordance with Drawing 13221E7392.

5.1.2 Lights procured as spares. Each light shall be preserved in accordance with MIL-P-116, method IA-15, using a box conforming to PPP-B-566. Each light shall be cushioned within the box with appropriate material or immobilized with fiberboard inserts or flipouts; the latter shall allow a minimum 1/2-inch clearance between the bulb and the sides, top, and bottom of the box.

5.1.2.1 Consolidation. Lights, preserved as specified in 5.1.2 shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c or W5c, style optional, in quantities not to exceed the size or weight limitation of the box.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

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5.2.1.1 Lights procured as sets for HEMMS. Lights procured as sets, preserved as specified in 5.1.1, shall be packed in a snug-fitting box conforming to PPP-B-601, overseas type, style 1. Not more than six sets of lights shall be packed in each box. The box shall have skids and shall be reinforced with flat steel strapping in accordance with the box specification.

5.2.1.2 Lights procured as spares. Lights procured as spares, preserved as specified in 5.1.2 and 5.1.2.1, shall be packed as specified in 5.2.1.1.

5.2.2 Level B. Lights, preserved as specified in 5.1.1 or 5.1.2 and 5.1.2.1, shall be packed in a snug fitting box conforming to PPP-B-636, grades V3c, V11c or V13c, style optional, in quantities not to exceed the size or weight limitations of the box. Box closure shall be in accordance with the appendix to the box specification.

5.2.3 Level C. Lights, preserved and consolidated as specified in 5.1.2.1, shall require no additional packing.

5.3 Unitization. Quantities of lights packed as level B or C and designated for shipment to single receiving points shall be palletized when specified (see 6.2). Pallets shall conform to NN-P-71, type and style optional, size 1 or 2, group II. The quantity and method of placement on the pallet shall be determined by the size and configuration of the boxes as recommended by the contractor. The load shall be secured to the pallet with four straps conforming to QQ-S-781, class 1, type I or IV, finish A, not less than 1/2 by 0.020 inches in size.

5.4 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The light is intended to be a component of HEMMS. As such, the lights are used to mark the perimeter and the safe lanes of minefields rapidly emplaced by friendly forces. The light, being completely self-contained, may be used for other appropriate purposes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Time frame required for submission of first article model(s) and number of lights required (see 3.5).

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d. When the Government will conduct any or all of the first article production model examinations and tests. When the Government will conduct some but not all of the first article examinations and tests, the contracting officer should specify which examinations and tests shall be conducted by the contractor (see 3.5 and 4.3).
e. When the contractor is required to perform the reliability test and reliability check test (see 4.4.3.11 and 4.4.3.11.3).
f. When the contractor is required to inspect a quantity of first article models other than stated in 4.4, the contracting officer should specify the quantity the contractor is to inspect.
g. Level of packing required (see 5.2).
h. When palletization is required (see 5.3).
i. When Government-furnished property is required (see 3.8.1 and 6.4).

6.3 First article. When a first article inspection is required, the items should be the initial production models. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Government-furnished property (GFP). The contracting officer should arrange to furnish the property specified in 3.8.1 and 3.12.1 when applicable. The GFP should be obtained by the contracting officer from the U.S. Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-NND, Fort Belvoir, VA 22060-5606.

6.5 Degree of preservation. Degrees of preservation have not been included, as the requirements specified are the minimum acceptable for any degree of protection.

6.6 Guidance for effective luminous measurements. The following Illuminating Engineering Society documents are recommended for calculating intensity measurements:

IES Guide for Calculating the Effective Intensity of Flashing Signal Lights, IES Transactions, November 1964.

Computation of the Effective Intensity of Flashing Lights, Charles A. Douglas, Illuminating Engineering, Vol. 52, No. 12, December 1957.

(Application for copies should be addressed to Illuminative Engineering Society, 345 East 47 Street, New York, NY 10017.)

6.7 Subject term (key word) listing.

Blinking
Light
Marker

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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activities:
Army - ER

Preparing activity:
Army - ME

Project 6220-A361

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-52998A(ME)	Light, Marker, Blinking

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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